Lane-Emden Equation Results

Symbolic regression / neural search experiments | n = 0, 1, 2, 3, 4, 5

1. Main Results Summary

n	Problem / Scope	Reward	Certified / Residual zero	Expression	Fitted expression (coeffs)	Max |err| interval	Success / Rate (if avail.)
0	lane_emden_n0_cleared scope=last_layer	26.000	residual_zero=True	a8 + a9/x - x^2/6	exact match to reference a0 + a1/x - x^2/6	-	59/64 92.188%
1	lane_emden_n1_cleared scope=last_layer	26.000	residual_zero=True	cos(a8 + x)/(a9 x + 2x)	exact (up to reparam.) (a0 sin x + a1 cos x)/x	-	61/64 95.312%
2	TTRL run	8.395	certified=True elite=True	a9 exp(-x cosh(log(a8 x + 2x)) + 1)	1.746 * exp(-x * cosh(log(0.321 * x)) + 1) a8=-1.679, a9=1.746	2.785e-2 on [0, 4]	-
3	TTRL run	7.984	certified=True elite=True	1/cosh(x + (a9*x*(a8 + x) + 2*x)/x)	1/cosh(x + (-1.567*x*(x + 1.279) + 2*x)/x) a8=1.279, a9=-1.567	3.785e-2 on [0.5, 6.8]	-
4	TTRL run	0.215	certified=False elite=True	1/(a9 + x*log(a8*x + 1))	1/(x*log(0.176*x + 1) + 1.000) a8=0.176, a9=1.000	4.868e-2 on [0.5, 14.8]	-
5	scope=last_layer	-4.563	residual_zero=False exact_IVP=False	a8*sin(a9 + x)/x	-	-	-
5	scope=decoder	-0.119	residual_zero=False exact_IVP=False	a8*sin(a9 + x/Abs(x))/x	-	-	-

Colour key: Green = exact recovery (residual_zero=True) Yellow = good approximate closed form Orange = weaker approximation Red = failed / negative reward

2. Absolute & Relative Errors - n = 2

Interval [0, 4] | max |error| = 2.785e-2

x	Abs. error	Rel. error
0.5	5.813e-4	6.059e-4
1.0	2.446e-3	2.882e-3
1.5	1.065e-3	1.531e-3
2.0	3.896e-3	7.353e-3
2.5	8.172e-3	2.181e-2
3.0	6.032e-3	2.494e-2
3.5	6.011e-3	4.487e-2
4.0	2.785e-2	5.703e-1

3. Absolute & Relative Errors - n = 3

Interval [0.5, 6.8] | max |error| = 3.785e-2

x	Abs. error	Rel. error
0.5	1.405e-4	1.464e-4
1.1	1.339e-3	1.601e-3
1.6	9.467e-5	1.395e-4
2.2	3.539e-3	6.709e-3
2.8	6.978e-3	1.747e-2

3.4	8.073e-3	2.717e-2
3.9	5.995e-3	2.762e-2
4.5	8.720e-4	5.656e-3
5.1	6.724e-3	6.443e-2
5.7	1.611e-2	2.507e-1
6.2	2.666e-2	8.475e-1
6.8	3.785e-2	9.082e+0

4. Absolute & Relative Errors - n = 4

Interval [0.5, 14.8] | max |error| = 4.868e-2

x	Abs. error	Rel. error
0.5	5.552e-4	5.782e-4
1.8	1.096e-3	1.642e-3
3.1	2.906e-4	6.829e-4
4.4	2.521e-3	8.959e-3
5.7	8.005e-3	4.133e-2
7.0	1.477e-2	1.083e-1
8.3	2.164e-2	2.245e-1
9.6	2.814e-2	4.190e-1
10.9	3.409e-2	7.604e-1
12.2	3.948e-2	1.448e+0
13.5	4.433e-2	3.387e+0
14.8	4.868e-2	3.499e+1

Notes: Reward = 26.000 with residual_zero = True indicates an exact symbolic solution was recovered (n=0,1). For n=2 and n=3 the search produced useful approximate closed forms (certified=True). n=4 is a weaker approximation (certified=False). n=5 did not recover a residual-zero solution under either last_layer or decoder scope; both final rewards are negative. First-zero error was reported as 1e6 (approx xi_1 = None) for n=2,3,4.